

DCN Finals Reviewer

Data Communication and Networking

How to Use This Reviewer

Exam-scope order

This reviewer is now grouped by the six exam scopes: **Network Layer**, **IPv4 Concepts**, **IPv6 Concepts**, **Transport Layer**, **Application Layer**, and **Network Security Fundamentals**. Each scope starts with a LaTeX-drawn mind map and a quick visual before the questions.

Acronyms to Memorize First

High-yield full names

- **DHCP** — Dynamic Host Configuration Protocol
- **DNS** — Domain Name System
- **FTP/TFTP** — File Transfer Protocol / Trivial File Transfer Protocol
- **HTTP/HTTPS** — Hypertext Transfer Protocol / Secure
- **SMTP, POP, IMAP** — Email send and retrieval protocols
- **SMB** — Server Message Block
- **URL** — Uniform Resource Locator
- **IP, IPv4, IPv6** — Internet Protocol versions
- **GUA, LLA, ULA** — Global, Link-Local, and Unique Local IPv6 addresses
- **RA/RS** — Router Advertisement / Router Solicitation
- **TCP/UDP** — Transmission Control Protocol / User Datagram Protocol
- **MSS, MTU, SACK, ISN** — Segment size, link size, selective ACK, initial sequence number
- **AAA** — Authentication, Authorization, Accounting
- **DMZ** — Demilitarized Zone
- **SPI** — Stateful Packet Inspection
- **IANA** — Internet Assigned Numbers Authority

Most Likely to Ask

Focus pass before answering drills

- **Network layer**: logical addressing, encapsulation, routing, fragmentation, best effort, connectionless delivery.
- **IPv4**: network/host portions, subnet mask, ANDing, private ranges, broadcast, loopback, NAT, VLSM.
- **IPv6**: 128-bit hex addresses, hextets, compression, GUA/LLA/ULA, ff02::1 and ff02::2 multicast, RA/RS, Hop Limit.
- **Transport**: TCP reliable/ordered/stateful; UDP connectionless/low overhead; ports identify application conversations.
- **Application**: HTTP methods, URL parts, DNS records, DHCP DORA, email protocols, FTP ports, P2P/client-server models.
- **Security**: threats, vulnerabilities, malware, attack types, firewall filtering, AAA, DMZ, password/SSH hardening.

1. Network Layer

LaTeX mind map

```
graph LR; NL[Network Layer] --> BO[Basic operations<br/>address end devices<br/>encapsulate packets<br/>route and de-encapsulate]; NL --> P[Packets<br/>IPv4 header limits<br/>MTU and fragmentation<br/>IPv6 extension headers]; NL --> HR[Host routing<br/>self route<br/>same LAN direct<br/>remote via gateway]; NL --> RT[Routing table<br/>route source<br/>best match<br/>default route]; NL --> IT[IP traits<br/>connectionless<br/>best effort<br/>media independent];
```

Cheat sheet / quick visual

```
graph LR; Host[Host] --> DQ{Destination?}; DQ --> SL[Same LAN<br/>send direct]; DQ --> RL[Remote LAN<br/>default gateway]; RL --> RT[Router routing table<br/>best match or default];
```

- **IP packet** = Network Layer PDU.
- **IPv4 fragmentation** happens when a packet is too large for the link MTU; IPv6 routers do not fragment packets.
- **Route source** tells how a router learned the route: connected, static, EIGRP, and so on.

Practice Questions

1. Which routes are automatically added by a router when an interface is active and has addressing?

Directly Connected

A directly connected route appears automatically for networks attached to active, addressed router interfaces.
2. Which route forwards traffic in a specific direction when there is no matching entry in the routing table?

Default Route

A default route is the fallback path used when a packet's destination does not match a more specific route.
3. What does it mean for IP to be **connectionless**?

IP does not establish a connection with the destination before sending the packet

Connectionless delivery means each packet is sent independently without first creating an end-to-end session.
4. Which type of route must be manually configured and adjusted by a network administrator?

Static Route

Static routes are entered by an administrator and do not automatically change when network conditions change.
5. What occurs when Layer 3 splits an IPv4 packet into smaller units?

Fragmentation

Fragmentation divides a packet into smaller pieces so it can fit across a link with a smaller maximum transmission unit.

6. Which code in a Cisco routing table indicates an EIGRP learned route?

D

Cisco routing tables use the code D for routes learned through EIGRP.
7. True or False: IP is considered unreliable because it cannot manage or fix undelivered or corrupt packets.

True

IP provides no built-in recovery for lost, delayed, duplicate, or corrupt packets; reliability is handled by upper-layer protocols when needed.
8. What represents the **Network Layer PDU**?

IP Packet

At the network layer, the protocol data unit is a packet, commonly an IP packet.
9. Which characteristic says IP does not guarantee delivery of a packet?

Best Effort

Best effort means IP attempts delivery but does not guarantee arrival, order, or error recovery.
10. In a router's routing table, what does the Route Source entry identify?

How the network was learned by the router

The route source code, such as C for connected or D for EIGRP, tells how the router learned the route.
11. What is the primary function of Read-only memory (ROM) in a Cisco router?

To provide startup instructions and perform the Power-on self-test (POST)

ROM contains essential boot microcode and hardware self-test instructions.

2. IPv4 Concepts

LaTeX mind map

Structure
32 bits
four octets
network plus host

Masking
subnet mask
prefix length
ANDing

Address types
network address
usable hosts
broadcast address

IPv4 Concepts

Delivery
unicast one-to-one
broadcast one-to-all
multicast one-to-group

Special ranges
private RFC1918
loopback 127/8
APIPA 169.254/16

Subnetting
borrow host bits
VLSM
match host need

Cheat sheet / quick visual

Example 192 . 168 . 10 . 15/24
Network bits first 24 bits = 192 . 168 . 10
Host bits last 8 bits = 15
Network address 192 . 168 . 10 . 0
Broadcast 192 . 168 . 10 . 255

192	168	10	15
network portion (/24)			host

Borrowing formula: borrowed bits n create 2^n subnets.

Practice Questions

12. How many total bits make up a standard **IPv4 address** structure?
- 32
- An IPv4 address is a 32-bit number, typically written as four decimal octets separated by dots (e.g., 192.168.1.1).
13. Which component is used by a network device to differentiate the **network portion** from the **host portion** of an IPv4 address?
- Subnet Mask
- The subnet mask is a 32-bit value where contiguous 1-bits mark the network portion and 0-bits mark the host portion of an address.
14. What is the logical result of the **ANDing process** when a binary 1 is compared to a binary 0?
- 0
- The AND operation returns 1 only when both bits are 1; any other combination (1 & 0, 0 & 1, 0 & 0) yields 0.
15. What is the **prefix length** for a subnet mask of **255.255.255.0**?
- /24
- Three octets of all 1s (8×3) equals 24 bits set in the mask.
16. According to **RFC 1918**, which of the following is a **private IPv4 address** range?
- 192.168.0.0/16
- This range is reserved for internal network use and is not globally routable.
17. What is sending a packet to all other destination IP addresses called?
- Broadcast
- Broadcast communication sends one packet to every host on the destination network.
18. What translates private IPv4 addresses to public IPv4 addresses?
- Network Address Translation (NAT)
- NAT lets private internal addresses communicate externally by translating them to public IPv4 addresses.
19. What are addresses commonly known as Automatic Private IP Addressing (APIPA) or self-assigned addresses?
- Link-Local Address
- IPv4 link-local/APIPA addresses are self-assigned when a host cannot obtain an address from DHCP.

20. Which address category is commonly identified by 127 . 0 . 0 . 1?
- Loopback Addresses
- Loopback addresses let a host send traffic to itself for testing and local services.
21. What is the bit value in the **Version** field for an IPv4 packet?
- 0100
- IPv4 is version 4, represented in the 4-bit Version field as 0100.
22. What two portions make up an IPv4 address?
- Network and host
- The network portion identifies the network, while the host portion identifies the specific device on that network.
23. Which process determines the network address from an IPv4 address and subnet mask?
- ANDing
- The IPv4 address and subnet mask are compared bit by bit; the result is the network address.
24. If you borrow 1 bit from the host portion of a /24 network, how many subnets are created?
- 2
- Borrowing n bits creates 2^n subnets; 2^1 equals 2.
25. What is the benefit of using Variable Length Subnet Masking (VLSM)?
- It allows subnets of different sizes and reduces unused host addresses
- VLSM lets each subnet match the needed host count instead of forcing every subnet to be the same size.
26. A host with IP address 192 . 168 . 10 . 15/24 belongs to which network address?
- 192.168.10.0
- With a /24 mask, the final octet is the host portion; setting it to 0 gives the network address.
27. What is the broadcast address for the network 172 . 16 . 0 . 0/16?
- 172.16.255.255
- A /16 leaves the final two octets for hosts; setting all host bits to 1 gives 255.255.
28. How many bits are borrowed to create 8 subnets?
- 3 bits
- 2^3 creates 8 subnets.
29. If an administrator needs to create 30 subnets, how many bits must be bor-

rowed from a /24 network?

5 bits

Borrowing 5 bits creates 32 subnets, the smallest power of 2 that can cover 30.

30. Which statement describes a slash /24 octet boundary?

The network portion ends after the third octet

Each octet is 8 bits, so 24 bits means the first three octets are the network portion.

31. What is the result of borrowing 2 bits from the third octet of a /8 network?

A /10 prefix

Adding 2 borrowed bits to the original /8 prefix produces /10.

32. A company uses 10.0.0.0/8 and needs subnets for 200 hosts each. Which

prefix length is most appropriate?

/24

A /24 subnet provides 254 usable host addresses, enough for 200 hosts with less waste than a larger subnet.

33. If you have a /16 network and need to create 256 subnets, what will be the new prefix length?

/24

Creating 256 subnets requires borrowing 8 bits; 16 + 8 = 24.

34. What is the theoretical maximum number of addresses provided by IPv4?

4.3 billion

IPv4 uses 32 bits, giving 2^{32} possible addresses, or about 4.3 billion.

3. IPv6 Concepts

LaTeX mind map

```
graph LR; IPv6[IPv6 Concepts] --> Why[Why IPv6<br/>IPv4 exhaustion<br/>128-bit space<br/>dual stack, tunneling,<br/>translation]; IPv6 --> Writing[Writing<br/>hex hextets<br/>omit leading zeros<br/>one double-colon run]; IPv6 --> Address[Address types<br/>unicast<br/>multicast<br/>anycast<br/>no broadcast]; IPv6 --> GUA[GUA<br/>global routing prefix<br/>subnet ID<br/>interface ID]; IPv6 --> LLA[LLA<br/>required per interface<br/>fe80::/10<br/>local link only]; IPv6 --> Dynamic[Dynamic addressing<br/>RS and RA<br/>SLAAC<br/>stateless or stateful<br/>DHCPv6];
```

Cheat sheet / quick visual

ff02::1 All-nodes multicast group: all IPv6 interfaces on the local link listen.
ff02::2 All-routers multicast group: IPv6 routers listen after routing is enabled.
fe80::/10 Link-local address range; stays on the local link.
2000::/3 Currently assigned Global Unicast Address range.

```
graph LR; Host[Host] --> RS[RS asks router]; RS --> RA[RA prefix + length]; RA --> SLAAC[SLAAC/DHCPv6 addressing];
```

Practice Questions

35. The devices run both **IPv4** and **IPv6** protocol stacks simultaneously.

Dual Stack

Dual stack allows a single device to process both IPv4 and IPv6 packets independently on the same interface.

36. A method of transporting an **IPv6 packet** over an **IPv4 network**. The IPv6 packet is **encapsulated** inside an IPv4 packet.

Tunneling

Tunneling wraps (encapsulates) an IPv6 packet inside an IPv4 header so it can traverse IPv4-only infrastructure without modification.

37. Which IPv6 transition mechanism lets IPv6-only devices communicate with IPv4-only devices by **converting packet headers** between the two protocols?

Translation

Translation converts packet headers between IPv6 and IPv4 formats, enabling direct communication between hosts running different IP versions.

38. Uniquely identifies an **interface** on an IPv6-enabled device.

Unicast

A unicast address identifies exactly one interface; packets sent to a unicast address are delivered to that single destination.

39. Used to send a single IPv6 packet to **multiple destinations**.

Multicast

Multicast delivers a packet to all members of a specific group, reducing bandwidth usage compared to sending individual copies to each recipient.

40. Any IPv6 unicast address that can be assigned to **multiple devices**. A packet sent to this address is routed to the **nearest device** having that address.

Anycast

Anycast uses routing metrics to forward the packet to the topologically closest interface sharing the same address, improving latency and redundancy.

41. Similar to a **public IPv4 address**. These are **globally unique, internet-routable** addresses.

Global Unicast Address (GUA)

GUAs are assigned from the 2000::/3 range and are equivalent to public IPv4 addresses, allowing devices to be reached across the global Internet.

42. Which IPv6 address type is required on every enabled interface, supports communication on the same **local link**, and is **not routable** beyond that link?

Link-Local Address (LLA)

Link-local addresses (fe80::/10) are automatically configured on every IPv6 interface and are used for neighbor discovery and routing protocol exchanges within a single network segment.

43. A **multicast group** that **all IPv6-enabled devices** join. A packet sent to this group is received and processed by all IPv6 interfaces on the link or network.

ff02::1 All-Nodes Multicast Group

ff02::1 is the link-local scope all-nodes multicast address, analogous to a broadcast in IPv4—every IPv6 host on the link listens to this address.

44. A **multicast group** that **all IPv6 routers** join. A router becomes a member of this group when it is enabled as an IPv6 router with the ipv6 unicast-routing global configuration command.

ff02::2 All-Routers Multicast Group

ff02::2 targets only routers on the local link; hosts send Router Solicitation messages to this address to discover available routers.

45. What is the total address length of an IPv6 address and written in what format?

128, Hexadecimal

IPv6 utilizes a 128-bit address space to provide a significantly larger number of unique addresses compared to IPv4.

46. The unofficial term used to refer to a segment of **16 bits**, or **four hexadecimal values**.

Hextet

A hextet refers to a four-hexadecimal digit segment representing 16 bits of the 128-bit address.

47. According to **Rule 1** of IPv6 compression, which zeros can be omitted?

Leading Zeros in Any Hextet

Only zeros at the start of a hextet can be omitted without making the address ambiguous.

48. Which of the following is a valid compression of the address 2001:0db8:0000:1111:0000:0000:0000:0200?

2001:db8:0:1111::200

This correctly applies Rule 1 (leading zeros) and Rule 2 (double colon for a contiguous string of all-zero hextets).

49. True or False: IPv6 addresses are **case-sensitive**.

False

IPv6 addresses are not case-sensitive and can be written in lowercase, uppercase, or a mix of both.

50. Which address type did IPv6 **eliminate** that was present in IPv4?

Broadcast

5

IPv6 does not have a broadcast address, using specific multicast groups instead to reduce unnecessary network noise.

51. What is the recommended **prefix length** for most IPv6 LANs?

/64

A 64-bit interface ID is required for SLAAC and simplifies network management and subnetting.

52. **Global Unicast Addresses (GUAs)** currently being assigned always begin with which **binary bits**?

001

The binary prefix '001' corresponds to the 2000::/3 address space currently used for GUAs.

53. Which field in a Global Unicast Address lies between the provider-assigned network portion and the host-identifying portion, allowing an organization to **identify internal subnets**?

Subnet ID

This field allows organizations to create internal subnets within their allocated global routing prefix.

54. Which field in a Global Unicast Address is the **network portion assigned by a provider**, such as an ISP, to a customer or site?

Global Routing Prefix

This portion forms the network identity that is routable on the global Internet.

55. Which field in an IPv6 address is equivalent to the **host portion** of an IPv4 address and is normally **64 bits long** on a /64 subnet?

Interface ID

The Interface ID uniquely identifies a host on a specific subnet.

56. What is the specific address range for **IPv6 Link-Local Addresses (LLAs)**?

fe80::/10

LLAs are required for every interface and always fall within the fe80::/10 range.

57. What prefix range is used by **Unique Local Addresses (ULAs)**?

fc00::/7 to fdff::/7

ULAs are used for local addressing within a site and fall within this specific range.

58. In the **EUI-64 process**, what value is inserted into the **middle of the 48-bit MAC address**?

fffe

The EUI-64 process takes the 48-bit MAC and inserts the 16-bit hex value 'fffe' in the middle to create a 64-bit ID.

59. Which **ICMPv6 message** is sent by a router to provide **addressing information** to hosts?

Router Advertisement (RA)

Routers send RA messages periodically or in response to an RS to provide prefix and gateway data.

60. How often do IPv6 routers typically send out **unsolicited ICMPv6 RA messages**?

Every 200 seconds

By default, IPv6-enabled routers send RA messages every 200 seconds to all nodes.

61. What is the restriction when applying IPv6 compression **Rule 2**, the double colon?

It can only be used once within an address

The double colon replaces one contiguous sequence of all-zero hextets; using it more than once would make the original address ambiguous.

62. What is the decimal equivalent of the address space provided by IPv6?

340 undecillion addresses

IPv6 provides 2¹²⁸ possible addresses, approximately 340 undecillion unique addresses.

63. Where are IPv6 Extension Headers (EH) placed in a packet?

Between the IPv6 header and the payload

Extension headers follow the main IPv6 header and appear before the transport-layer payload.

64. Which IPv6 header field indicates the size of the data portion of the packet?

Payload Length

Payload Length specifies the size of the data following the main IPv6 header.

65. Which organization developed IPv6?

Internet Engineering Task Force (IETF)

IPv6 was developed by the IETF to address IPv4 exhaustion and modernize IP addressing.

66. What is the binary representation of the **Version** field in an IPv6 header?

0110

IPv6 is version 6, represented in the 4-bit Version field as 0110.

67. In what hex range do all currently assigned IPv6 Global Unicast Addresses (GUAs) fall?

2000::/3

Currently assigned GUAs begin with binary 001, which corresponds to addresses in 2000::/3.

68. If an IPv6-enabled interface is not manually configured with an LLA, what will the device do?

It will automatically create one

IPv6 requires a link-local address on enabled interfaces, so devices can generate one automatically.

69. Which ICMPv6 message is sent by a host to discover IPv6 routers on the network?

Router Solicitation (RS)

A host sends an RS message to ask routers to provide configuration information immediately.

70. What does a Router Advertisement (RA) message typically provide to a host?

Network prefix and prefix length

RA messages provide the prefix information a host can use for SLAAC or related IPv6 configuration.

71. Which IPv6 header field is the equivalent of the IPv4 Time-to-Live (TTL) field?

Hop Limit

Hop Limit is decremented at each hop to prevent packets from looping forever.

72. How does the size of an IPv6 header compare to a standard IPv4 header?

The IPv6 header is larger, 40 bytes versus 20 bytes

IPv6 has a simplified fixed header, but its 128-bit addresses make the base header 40 bytes.

4. Transport Layer

LaTeX mind map

A mind map with 'Transport Layer' at the center. It branches into four categories: 'Role' (app-to-app delivery, segmentation, multiplexing), 'UDP' (connectionless, best effort, low overhead), 'Ports' (socket = IP + port, well-known, registered, dynamic), and 'TCP' (connection-oriented, reliable and ordered, flow control). There is also a 'TCP process' box listing SYN, SYN-ACK, ACK, sequence and ACK, and FIN termination.

Cheat sheet / quick visual

A diagram showing the TCP three-way handshake between a Client and a Server. Step 1: Client sends 1 SYN to Server. Step 2: Server sends 2 SYN + ACK to Client. Step 3: Client sends 3 ACK to Server.

TCP	reliable, ordered, flow control, retransmission, higher overhead
UDP	low delay, low overhead, no built-in retransmission
Ports	0–1023 well-known; 1024–49151 registered; 49152–65535 dynamic

Practice Questions

73. Responsible for establishing a temporary communication session between two applications and delivering data between them.

Transport Layer

The transport layer manages end-to-end communication by establishing sessions and ensuring data reaches the correct application.

74. Which transport layer protocol is considered 'reliable' because it supports packet delivery confirmation?

TCP

TCP uses sequence numbers and acknowledgments to ensure all data is delivered and processed.

75. What is the overhead size of a standard TCP header?

20 bytes

The TCP header contains several fields for reliability and flow control, totaling 20 bytes of overhead.

76. Which TCP flag is used to abruptly terminate a session if necessary?

RST

The Reset (RST) flag is used to immediately close a connection without the standard handshake.

77. Which is a 'Stateless' protocol?

UDP

UDP does not track session data or confirm delivery, making it stateless.

78. When using netstat -n, what does the '-n' option do?

Displays IP addresses and port numbers in numerical form.

This prevents the utility from attempting to resolve addresses to names, speeding up the output.

79. In a TCP session termination, which flag is used by the client to indicate it has no more data to send?

FIN

The Finish (FIN) flag signals the end of data transmission from that side of the connection.

80. What is the primary role of the 'Destination Port' in a header?

To identify the service or application running on the server.

Ports like 80 or 443 tell the receiving OS which software should handle the incoming data.

81. Which layer is responsible for logical communications between applications running on different hosts?

Transport Layer

The transport layer provides end-to-end logical communication between application processes on different devices.

82. Which layer is the link between the application layer and the lower layers responsible for network transmission?

Transport Layer

The transport layer accepts application data, segments it, and passes it to the lower layers for network delivery.

83. Which characteristic is a hallmark of the User Datagram Protocol (UDP)?

Connectionless Best-Effort Delivery

UDP does not establish a formal session before sending data and does not provide acknowledgments for received packets.

84. Which TCP flag indicates that the urgent pointer field is significant?

URG

The URG flag tells the receiver that the urgent pointer field should be interpreted as urgent data information.

85. Which TCP flag is the acknowledgment flag used in connection establishment and session termination?

ACK

The ACK flag confirms that data or control information has been received and identifies the next expected byte.

86. Which TCP flag represents the push function?

PSH

The PSH flag asks the receiver to pass buffered data to the application as soon as possible.

87. Which TCP flag resets the connection when an error or timeout occurs?

RST

The RST flag abruptly terminates a TCP connection when the session cannot continue normally.

88. Which TCP flag synchronizes sequence numbers during connection establishment?

SYN

The SYN flag starts a TCP session by synchronizing initial sequence numbers between endpoints.

89. Which TCP flag means no more data from the sender and is used in session termination?

FIN

The FIN flag tells the other endpoint that the sender has finished transmitting data.

90. What is the **maximum amount of data** that the destination device can receive in a TCP segment called?

Maximum Segment Size (MSS)

MSS defines the largest TCP payload that can be sent in one segment without including the IP and TCP headers.

91. If an IPv4 packet has an Ethernet **MTU of 1500 bytes**, what is the typical TCP Maximum Segment Size for data?

1460 bytes

Subtracting the 20-byte IPv4 header and the 20-byte TCP header from the 1500-byte MTU leaves 1460 bytes for TCP data.

92. What is the purpose of **Selective Acknowledgment (SACK)** in TCP?

To explicitly acknowledge which segments were received

SACK lets the receiver identify received and missing byte ranges, so the sender retransmits only the missing pieces.

93. True or False: UDP is considered a **stateful protocol** because it monitors the status of the connection.

False

UDP is stateless; it sends data without tracking a session or checking whether the destination is ready to receive.

94. During TCP session termination, how many steps are typically involved to close the connection in both directions?

Four

TCP termination normally uses a FIN and ACK from one side, followed by a FIN and ACK from the other side.

95. Which TCP header field is a **16-bit field** used to identify the source application by port number?

Source Port

The source port identifies the sending application or process on the source host.

96. Which TCP header field is a **16-bit field** used to identify the destination application by port number?

Destination Port

The destination port identifies the service or application that should receive the segment.

97. Which TCP header field is a **32-bit field** used for data reassembly purposes?

Sequence Number

The sequence number identifies the order of bytes so TCP can reassemble data correctly.

98. Which TCP header field is a **32-bit field** used to indicate that data has been received and to identify the next byte expected?

Acknowledgment Number

The acknowledgment number confirms receipt and tells the sender which byte the receiver expects next.

99. Which TCP header field is a **4-bit field** also known as the data offset?

Header Length

Header Length indicates the size of the TCP segment header so the receiver can locate the data payload.

100. Which TCP header field is a **6-bit field** reserved for future use?

Reserved

The reserved field is kept for future TCP extensions and should normally be set to zero.

101. Which TCP header field contains bit codes or flags that indicate the **purpose and function** of the TCP segment?

Control Bits

Control bits include flags such as URG, ACK, PSH, RST, SYN, and FIN.

102. Which TCP header field is a **16-bit field** that indicates how many bytes can be accepted at one time?

Window Size

Window Size supports flow control by advertising how much data the receiver can accept.

103. Which TCP header field is a **16-bit field** used for error checking of the segment header and data?

Checksum

The checksum helps detect corruption in the TCP header and data.

104. Which TCP header field is a **16-bit field** used to indicate if the contained data is urgent?

Urgent Pointer

The urgent pointer is meaningful when the URG flag is set and points to urgent data in the segment.

105. What is the combination of an **IP address and port number** known as?

Socket

A socket identifies an endpoint by combining an IP address with a transport-layer port number.

106. Which protocol is best suited for applications that tolerate some data loss but cannot tolerate delays, such as live audio streaming?

UDP

UDP avoids retransmission and connection overhead, making it useful for time-sensitive traffic where speed matters more than perfect delivery.

107. What congestion behavior results in packets being **discarded by an overloaded router**?

Congestion Avoidance

When congestion avoidance mechanisms detect overload, excess packets may be dropped to protect the network from collapse.

108. What endpoint identifier is used to identify the **server and service** being requested by a client?

Socket

The destination socket combines the server IP address and destination port for the requested service.

109. What is the IANA range for **well-known ports**?

0 to 1023

Ports 0 through 1023 are reserved for common services such as HTTP, FTP, DNS, and SSH.

110. Which network utility can verify **active connections** in a host?

netstat

netstat displays active network connections, routing tables, and interface statistics.

111. Which port number is typically associated with **HTTP**?

80

Port 80 is the well-known port for standard unencrypted HTTP web traffic.

112. What is the **Initial Sequence Number (ISN)** in a TCP connection?

A randomly chosen sequence number used during TCP session setup

The ISN is chosen during connection setup and then incremented by the number of transmitted bytes.

113. True or False: A single server can host both a web service on port 80 and an FTP service on port 21 at the same time.

True

A server can run multiple services simultaneously because each service listens on a different port number.

114. Which standards body assigns addressing standards, including port numbers?

Internet Assigned Numbers Authority (IANA)

IANA manages global coordination of DNS root data, IP addressing, and protocol resources such as port numbers.

115. Which flag is used in the first step of the TCP session termination process?

FIN

The FIN flag begins graceful session termination by indicating that the sender has no more data to transmit.

116. Which protocol is a better choice for an application that requires all data to be fully received before any of it is useful, such as a software update?

TCP

TCP provides reliability, ordering, acknowledgments, and retransmission, which are required for complete file delivery.

117. TCP and UDP both use which layer of the TCP/IP model?

Transport

TCP and UDP are transport-layer protocols that manage end-to-end delivery between applications.

118. What port number range is used for Registered Ports as assigned by IANA?

1,024 to 49,151

Registered ports are assigned to requesting entities for specific applications and services.

119. Which well-known port is used for the control connection of FTP sessions?

21

FTP uses port 21 for control commands and port 20 for data transfer in active FTP.

120. How many bits are used for the Source Port and Destination Port fields in both TCP and UDP headers?

16 bits each

Both TCP and UDP use 16-bit source and destination port fields, allowing port numbers from 0 to 65,535.

121. Which feature is unique to TCP compared to UDP?

Flow control

TCP uses flow control to prevent the sender from overwhelming the receiver; UDP does not.

122. What is the total size of a standard UDP header?

8 bytes

The UDP header has four 16-bit fields: source port, destination port, length, and checksum.

123. Which application layer protocol uses UDP port 69 and requires reliability to be handled by the application itself?

TFTP

Trivial File Transfer Protocol uses UDP port 69, so the application must handle reliability behavior.

124. Which protocol is connection-oriented and supports same-order delivery?

TCP

TCP establishes a session and uses sequence numbers so data can be delivered reliably and in order.

125. What is the primary advantage of using UDP for live video and multimedia applications?

It provides minimal delay by not retransmitting lost data or waiting for acknowledgments

Real-time media usually values low latency more than perfect retransmission.

126. Which port number is used by the Simple Mail Transfer Protocol (SMTP)?

25

SMTP commonly uses well-known port 25 for mail transfer.

127. What happens during Step 2 of the TCP three-way handshake?

The server responds with SYN and ACK flags

The server acknowledges the client SYN and sends its own SYN to synchronize sequence numbers.

128. What port range is typically used by a client's operating system as ephemeral ports?

49,152 to 65,535

Private or dynamic ports identify temporary client-side conversations.

129. Which port number is used by the DHCP server to listen for requests?

67

DHCP servers listen on UDP port 67, while clients use UDP port 68.

5. Application Layer

LaTeX mind map

```
graph LR; AL[Application Layer] --> UR[Upper-layer roles<br/>application: user services<br/>presentation: format, compress, encrypt<br/>session: dialogs]; AL --> M[Models<br/>client-server<br/>peer-to-peer<br/>hybrid P2P]; AL --> W[Web<br/>HTTP and HTTPS<br/>GET, POST, PUT<br/>URL parts]; AL --> E[Email<br/>SMTP sends<br/>POP retrieves<br/>IMAP syncs]; AL --> S[Services<br/>DNS records<br/>DHCP DORA<br/>nslookup]; AL --> FS[File sharing<br/>FTP control and data<br/>TFTP UDP 69<br/>SMB file and print];
```

Cheat sheet / quick visual

Discover

→

Offer

→

Request

→

ACK

URL	http://www.cisco.com/index.html = protocol + server name + filename
SMTP	sends mail, port 25
POP	downloads/retrieves, commonly removes from server
IMAP	synchronizes mail while originals remain on server
DNS records	A = IPv4, AAAA = IPv6, NS = name server, MX = mail exchange
FTP	pull = download; control port 21, data port 20

Practice Questions

130. The layer that is closest to the **end user**.
Application Layer
The Application layer is the topmost layer of both the OSI and TCP/IP models, providing the interface through which users interact with network services such as HTTP, FTP, and DNS.
131. A client/server file sharing protocol.
Server Message Block (SMB)
132. What are the 3 primary functions of the presentation layer?
Format data, Compress data, Encrypt data
133. Another protocol used to retrieve email messages.
IMAP
134. Allows the user to manually place DNS queries.
nslookup
135. Which protocol is a 'request/response' protocol that specifies GET, POST, and PUT message types and is not secure. Messages can be intercepted.
HTTP
The Hypertext Transfer Protocol uses these specific request methods to interact with resources on a web server.
136. When a DHCP lease must be renewed, which message does the client send?
DHCPREQUEST
A client sends a REQUEST message to ask for an extension of its existing lease before it expires.
137. Which message type is sent by a DHCP server if an IP address offer is no longer valid?
DHCPNAK
A negative acknowledgment (NAK) tells the client that the server cannot fulfill the specific request.
138. Which layer of the OSI model is responsible for providing the **interface between user applications and the network**?
Application Layer
The Application layer is closest to the end user and serves as the primary interface for network-aware applications.
139. Which layer of the OSI model is responsible for creating and maintaining **dialogs** between source and destination applications?
Session Layer

- The Session layer manages the exchange of information to keep conversations active and restart them if necessary.*
140. What is the primary role of a **tracker** in BitTorrent technology?
To keep track of which files are hosted by which users
Trackers maintain records of which peers have specific file segments so other peers can locate and share them.
141. Which layer of the OSI model formats data at the source device into a format compatible with the destination device?
Presentation Layer
The Presentation layer helps ensure that data sent from the Application layer of one system can be read by the Application layer of another.
142. Which DNS resource record type maps a domain name to an **IPv6 address**?
AAAA
The quad-A record is designed for 128-bit IPv6 addresses.
143. Which DHCP message is broadcast by a client to find available servers when it first connects to a network?
DHCPDISCOVER
The client uses DHCPDISCOVER to locate DHCP servers on the local network segment.
144. Which HTTP method is a client request for data, such as when a web browser requests an HTML page?
GET
GET asks the web server to return a requested resource.
145. Which HTTP method uploads data files or form data to a web server?
POST
POST sends data to a server for processing, such as submitted form contents.
146. Which HTTP method uploads or replaces resources or content on a web server, such as an image?
PUT
PUT sends a resource to the server, commonly to create or replace content at a specified location.
147. Which graphic image format mentioned in the material is a common standard handled at the Presentation layer?
PNG
Portable Network Graphics (PNG) is a common image format handled as presentation-layer data formatting.

148. What is a distinguishing characteristic of a Peer-to-Peer (P2P) network model?

Every connected device can act as both a client and a server

P2P roles are decentralized; a peer can request a resource in one transaction and provide one in another.
149. How does a hybrid P2P system differ from a standard P2P system?

An index server helps peers locate resources while resource sharing remains decentralized

Hybrid P2P keeps peer-to-peer file sharing but uses a central index or tracker to help peers find who has the resource.
150. What are the three basic parts of a URL such as protocol, server name, and filename?

Protocol, server name, and specific filename

A URL identifies the scheme or protocol, the host or server name, and the resource or file being requested.
151. What is the primary difference between HTTP and HTTPS?

HTTPS uses authentication and encryption to secure data

HTTP sends data without encryption; HTTPS protects the session so intercepted traffic cannot be read easily.
152. A store-and-forward method of sending, storing, and retrieving electronic messages.

Email

Email stores messages on servers until the recipient retrieves or synchronizes them.
153. What is a major advantage of using IMAP over POP for retrieving email?

It lets users view copies of messages while the originals remain on the server

IMAP keeps mail on the server and synchronizes state, which is useful when the same mailbox is opened from many devices.
154. Which DNS record type is used to identify an authoritative name server?

NS

An NS record names the authoritative DNS servers for a zone.
155. In the DNS hierarchy, which top-level domain represents a non-profit organization?

.org

The .org top-level domain is commonly associated with non-profit organizations.
156. What happens to a DHCP-distributed IP address when it is no longer in use?

It is returned to the address pool for reuse

- DHCP leases allow unused addresses to return to the pool so another host can receive them later.
157. Which protocol is primarily used for file-sharing and print services in Windows-based networking?

Server Message Block (SMB)

SMB is the common Windows resource-sharing protocol for files, printers, and related services.
158. What is the function of the DNS MX record?

A mail exchange record identifies where email for a domain should be delivered

MX records point sending mail servers to the mail exchangers responsible for a domain.
159. What Windows command displays all cached DNS entries?

ipconfig /displaydns

The /displaydns option shows the local DNS resolver cache.
160. What is the function of the DNS A record?

It maps a name to an end-device IPv4 address

An A record links a human-readable host name to a 32-bit IPv4 address.
161. Which common video file standard is processed at the Presentation layer?

MPEG

MPEG is a common video standard, so it fits the presentation-layer role of formatting data.
162. What port number does an SMTP client typically use to connect to a mail server?

25

SMTP uses well-known port 25 to transfer email between mail systems.
163. How does an FTP client pull data from a server?

By downloading a file

Pulling data means the client requests and receives a file from the server.
164. In the DNS hierarchy, what do top-level domains like .au or .co usually represent?

Country-code top-level domains

Two-letter TLDs are usually country-code top-level domains assigned to countries or territories.
165. True or False: Application layer protocols must be implemented in both the source and destination devices to allow communication.

True

Both endpoints must use compatible protocol rules to understand the same application-layer exchange.

6. Network Security Fundamentals

LaTeX mind map

Threats

information theft

data loss

identity theft

service disruption

Attacks

reconnaissance

access

DoS and DDoS

MitM

Vulnerabilities

technology

configuration

policy

physical

Mitigation

defense-in-depth

AAA

DMZ

firewalls

Malware

virus: host file

worm: standalone

trojan: looks legitimate

Hardening

strong passwords

SSH and RSA

login block-for

disable unused services

Network Security

Cheat sheet / quick visual

Authentication

Who are you?

Authorization

What are you allowed to do?

Accounting

What did you do?

Packet filtering

IP/MAC allow or block rules

Application filtering

allows or blocks by port/application type

URL filtering

allows or blocks by URLs or keywords

SPI

permits legitimate replies, blocks unsolicited packets

DMZ

controlled public-service subnet separated from the inside network

Practice Questions

166. The **discovery and mapping** of systems, services, or vulnerabilities.

Reconnaissance Attacks

Reconnaissance is the information-gathering phase where attackers probe networks using tools like ping sweeps and port scans to identify potential targets.

167. The unauthorized **manipulation of data**, system access, or **user privileges**.

Access Attacks

Access attacks exploit authentication weaknesses or software vulnerabilities to gain unauthorized entry to systems or escalate privileges.

168. The **disabling or corruption** of networks, systems, or services.

Denial of Service

DoS attacks flood a target with traffic or exploit protocol weaknesses to exhaust resources, rendering the service unavailable to legitimate users.

169. A type of malware that propagates by **inserting a copy of itself** into, and becoming part of, another program. It spreads from one computer to another, leaving infections as it travels.

Viruses

Viruses require a host file to propagate; they attach to legitimate programs and execute when the host program runs.

170. Similar to viruses in that they **replicate functional copies** of themselves and can cause the same type of damage. In contrast to viruses, which require the spreading of an infected host file, this is a **standalone software** and does not require a **host program** or human help to propagate.

Worms

Worms are self-propagating and exploit network vulnerabilities to spread automatically without user interaction or a host executable.

171. A harmful piece of software that **looks legitimate**. They do not reproduce by infecting other files. They **self-replicate** and spread through **user interaction** such as opening an email attachment or downloading and running a file from the internet.

Trojan Horses

Trojans disguise themselves as useful software to trick users into installing them, after which they can create backdoors, steal data, or download additional malware.

172. Malware is short for...

Malicious Software

"Malware" is the umbrella term for any software intentionally designed to cause damage, including viruses, worms, trojans, ransomware, and spyware.

173. Includes **physical damage** to servers, routers, switches, cabling plant, and workstations.

Hardware Threats

Hardware threats encompass any event—accidental or deliberate—that physically damages networking equipment, such as vandalism, theft, or natural disasters.

174. Includes **temperature extremes** (too hot or too cold) or **humidity extremes** (too wet or too dry).

Environmental Threats

Environmental threats affect equipment reliability; excessive heat causes overheating, while high humidity can lead to condensation and corrosion on circuit boards.

175. Includes **voltage spikes**, insufficient supply voltage (**brownouts**), unconditioned power (**noise**), and **total power loss**.

Electrical Threats

Electrical threats are mitigated by using UPS (uninterruptible power supplies), surge protectors, and power conditioners to maintain clean, stable electricity.

176. Includes **poor handling** of key electrical components (**electrostatic discharge**), lack of critical spare parts, poor cabling, and poor labeling.

Maintenance Threats

Maintenance threats arise from neglecting best practices such as ESD wrist straps, proper cable management, and keeping an adequate inventory of spare parts.

177. Implemented using **brute force**, **Trojan horse**, and **packet sniffers**.

Password Attacks

Password attacks attempt to discover credentials through exhaustive guessing (brute force), keystroke logging (Trojans), or intercepting unencrypted credentials on the network (sniffers).

178. A threat actor uses **unauthorized privileges** to gain access to a system, possibly compromising the target.

Trust Exploitation

Trust exploitation abuses the trust relationships between systems—e.g., if System A trusts System B, compromising B gives the attacker access to A.

179. A threat actor uses a **compromised system** as a base for attacks against other

14

targets. For example, using **SSH (port 22)** to connect to a compromised host A. Host A is trusted by host B and, therefore, the threat actor can use **Telnet (port 23)** to access it.

Port Redirection

Port redirection (pivoting) allows an attacker to tunnel traffic through a compromised host to reach otherwise inaccessible systems on the internal network.

180. The threat actor is positioned **in between two legitimate entities** in order to **read or modify** the data that passes between the two parties.

Man-in-the-Middle

MitM attacks intercept communications in real time, enabling eaves-dropping, session hijacking, or data manipulation without either party’s knowledge.

181. Prevents or allows access based on **IP or MAC addresses**.

Packet Filtering

Packet filtering examines the source and destination addresses in each packet header and applies allow/deny rules without inspecting payload content.

182. Prevents or allows access by specific **application types** based on **port num-bers**.

Application Filtering

Application filtering controls traffic by matching well-known port numbers (e.g., port 80 for HTTP, port 443 for HTTPS) to allow or block specific services.

183. Prevents or allows access to websites based on specific **URLs or keywords**.

URL Filtering

URL filtering inspects the requested web address or embedded keywords and blocks access to sites that match a deny list or policy category.

184. Which firewall filtering method tracks active connections, permits incoming packets that are **legitimate responses** to internal requests, and blocks **unsolicited packets** unless explicitly allowed?

Stateful Packet Inspection (SPI)

SPI maintains a state table of active connections and only permits inbound traffic that corresponds to an established outbound session, providing stronger security than simple packet filtering.

185. Which term describes intruders who gain access by modifying software or exploiting software vulnerabilities?

Threat actors

Threat actors are individuals or groups that exploit weaknesses to gain unauthorized access or cause harm.

186. A network suffers from a weakness in the TCP/IP protocol itself. Which vul-nerability category is this?

Technological vulnerabilities

Protocol, operating system, and hardware weaknesses are technological vulnerabilities.

187. Which item is considered a configuration vulnerability?

Unsecured user accounts

Weak or default account settings are configuration problems.

188. Which malware is a standalone program that replicates itself and does not need a host program to spread?

Worm

A worm is independent software that can replicate across networks without attaching to a host file.

189. Which tool might a threat actor use during reconnaissance to determine the IP address space assigned to a corporation?

nslookup

Tools such as nslookup and whois help gather DNS, domain, and address information.

190. What is the term for infected hosts controlled by a command-and-control program to perform attacks?

Botnet

A botnet is a group of compromised hosts controlled together for attacks such as DDoS.

191. Which security approach combines networking devices and services in a lay-ered fashion?

Defense-in-depth

Defense-in-depth layers controls so another protection can help when one control fails.

192. Which part of AAA determines exactly what a user is allowed to do while logged into the network?

Authorization

Authorization defines permitted actions and resources after identity is verified.

193. Which AAA component keeps a record of what the user did while accessing the network?

Accounting

Accounting logs user activity for audits, tracking, and accountability.

194. Where are servers accessible to outside users usually placed to allow con-trolled access without compromising the internal network?

Demilitarized Zone (DMZ)

A DMZ exposes public services in a controlled subnet while separating them from the internal network.

195. Why is securing endpoint devices one of the most challenging jobs for a net-work administrator?

It involves human nature and user behavior

Endpoint security depends on technical controls and on users following safe practices.

196. What is the minimum recommended length for a strong password?

Eight characters

Eight characters is a common minimum, while longer passwords or passphrases are stronger.

197. What is a passphrase in Cisco device security?

A phrase made of many words that is longer and harder to guess

A passphrase uses multiple words to create something long, memorable, and more resistant to guessing.

198. Which Cisco command prevents plaintext passwords from being visible in the configuration file?

service password-encryption

This command encrypts plaintext passwords in the running configuration.

199. What is the purpose of the command `login block-for 120 attempts 3 within 60`?

To deter brute-force password guessing attacks

The command blocks login attempts after repeated failures within a short period.

200. What is the minimum recommended RSA key modulus length for secure SSH configuration?

1024 bits

Cisco guidance commonly treats 1024 bits as the minimum practical RSA key size for SSH.

201. In Cisco IOS-XE, which command verifies which ports are currently open on the device?

show ip ports all

This command displays open IP ports on IOS-XE devices.